

Homework 3*Due Friday, March 20*

1. Let Δ be a Cohen-Macaulay simplicial complex and Δ' a Cohen-Macaulay subcomplex of Δ . Assume Δ' is an *induced subcomplex* of Δ ; that is, if $\sigma \in \Delta$ and all vertices of σ are in Δ' , then $\sigma \in \Delta'$. Prove that $h_i(\Delta') \leq h_i(\Delta)$ for all i .
2. Show that the h -vector of a Cohen-Macaulay complex is not always unimodal. (Note: Every connected shellable complex is Cohen-Macaulay.)